

Generation Interconnection System Impact Study Report

For Queue Project Q15-0482

Garden 138 kV

May 21, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0482. The Interconnection Customer proposes a 133.4 MW solar photovoltaic facility interconnecting at the Garden 138 kV substation in the Virginia area. The request was received on May 30, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 285 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$11,865,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
WESTROCK SOUTHEAST LLC - PACIFICA 138 KV	MITCHELL DAM - SKAGGS ISLAND JCT 2 345 KV	182.8	166	6.7
NATCHEZ SUBSTATION - CHANCELLOR ROW 138 KV	MITCHELL DAM - SKAGGS ISLAND JCT 2 SUBSTATION 345 KV	178.9	213	8.4
MITCHELL DAM SUBSTATION - SYL LASKIN 138 KV	MITCHELL DAM - HAGGIN 345 KV	176.8	228	7.1
CARVER - BIG HANAFORD 138 KV	PLAZA - BIG HANAFORD 138 KV	175.0	187	6.4
DELTPORT - MOONSTONE 138 KV	MITCHELL DAM - SKAGGS ISLAND JCT 2 345 KV	173.0	222	7.1
N TOWER TAP - CEDAR FALLS WI 138 KV	NORTH TOWER TAP - HAGGIN 345 KV	171.3	304	18.5
MCCARTNEY - N TOWER TAP 345 KV	KERR DAM - TECUMSEH ENERGY CTR 345 KV L	170.4	1,435	6.9
CARVER - SHOCKOE 138 KV	PLAZA - BIG HANAFORD 138 KV	170.0	230	6.4
POTOSI - DELTAPORT 138 KV	SKAGGS ISLAND JCT 2 - MITCHELL DAM 345 KV	169.5	246	7.1
N TOWER TAP - HAGGIN 345 KV Line	KERR DAM - TECUMSEH ENERGY CTR 345 KV	169.2	1,281	26.7
HARVELL - OYSTER RIV 138 KV	N TOWER TAP - HAGGIN 345 KV L	167.6	191	10.8
JIM HILL - HAGIN 138 KV	SKAGGS ISLAND JUNCTION 2 - MITCHELL DAM SUBSTATION 345 KV	166.9	257	7.7
SPRING ST - CHIPOLA 138 KV	SKAGGS ISLAND JCT 2 SUBSTATION - MITCHELL DAM 345 KV line	166.3	234	8.4

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
WESTCOST CELLUFIBRE - CHIPOLA 138 KV	SKAGGS ISLAND JCT 2 - MITCHELL DAM 345 KV	166.0	257	11.9
POTOSI SUBSTATION - TYLER 138 KV Line	MITCHELL DAM - SKAGGS ISLAND JCT 2 345 KV	165.0	269	7.1
CENTRALIA - CEDAR BAYOU PLANT 138 KV line	MITCHELL DAM - SKAGGS ISLAND JUNCTION 2 345 KV	161.0	208	7.7
HIGHLAND CITY - CENTRALIA 138 KV line	SKAGGS ISLAND JCT 2 - MITCHELL DAM SUBSTATION 345 KV	160.8	245	6.7
CEDAR FALLS WI - OYSTER RIV 138 KV	NORTH TOWER TAP - HAGGIN 345 KV L	160.3	212	10.8
CEDAR BAYOU PLANT - JIM HILL 138 KV L	MITCHELL DAM - SKAGGS ISLAND JCT 2 345 KV line	159.4	236	7.7
NORTH RANDOLPH - YELLOW CRK 138 KV Line	HAGGIN - HULL STREET RD 345 KV line	158.1	241	7.0
MUKONAGO - PACIFICA 138 KV	SKAGGS ISLAND JCT 2 - MITCHELL DAM 345 KV L	157.1	222	6.7
HIGHLAND CITY - MUKONAGO 138 KV	MITCHELL DAM - SKAGGS ISLAND JUNCTION 2 345 KV	156.6	235	6.7
N RANDOLPH - BAIROA 3008 138 KV	HULL ST ROAD - HAGGIN 345 KV L	155.7	237	7.0
WALTHALL - HAGIN 138 KV	N TOWER TAP - HAGGIN 345 KV	154.9	439	24.4
SKAGGS ISLAND JCT 2 - MITCHELL DAM 345 KV	BEAR - MECKLENBURG POWER STATION 345 KV line	154.6	1,191	28.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
PACIFICA - WESTROCK SOUTHEAST LLC 138 KV line	Rebuild line with higher-capacity conductor	\$7,800,000	7% (\$574,000)
NATCHEZ SUBSTATION - CHANCELLOR ROW 138 KV	Rebuild line with higher-capacity conductor	\$2,404,000	3% (\$69,000)
MITCHELL DAM - SYL LASKIN 138 KV	Rebuild line with higher-capacity conductor	\$1,563,000	2% (\$38,000)
CARVER - BIG HANAFORD 138 KV	Rebuild line with higher-capacity conductor	\$1,679,000	6% (\$103,000)
MOONSTONE - DELTAPORT 138 KV line	Rebuild line with higher-capacity conductor	\$3,006,000	3% (\$87,000)
N TOWER TAP - CEDAR FALLS WI 138 KV	Rebuild line with higher-capacity conductor	\$9,573,000	15% (\$1,452,000)
NORTH TOWER TAP - MCCARTNEY 345 KV	Rebuild line with higher-capacity conductor	\$87,810,000	0% (\$441,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CARVER - SHOCKOE 138 KV	Rebuild line with higher-capacity conductor	\$1,245,000	6% (\$77,000)
POTOSI - DELTAPORT 138 KV line	Rebuild line with higher-capacity conductor	\$4,938,000	3% (\$143,000)
N TOWER TAP - HAGGIN 345 KV Line	Rebuild line with higher-capacity conductor	\$56,110,000	2% (\$1,259,000)
HARVELL - OYSTER RIV 138 KV	Rebuild line with higher-capacity conductor	\$1,707,000	10% (\$165,000)
JIM HILL - HAGGIN 138 KV	Rebuild line with higher-capacity conductor	\$1,007,000	3% (\$29,000)
SPRING ST - CHIPOLA 138 KV	Rebuild line with higher-capacity conductor	\$19,396,000	4% (\$675,000)
CHIPOLA - WESTCOAST CELLUFIBRE 138 KV	Rebuild line with higher-capacity conductor	\$1,989,000	3% (\$60,000)
POTOSI SUBSTATION - TYLER 138 KV Line	Rebuild line with higher-capacity conductor	\$700,000	3% (\$20,000)
CENTRALIA - CEDAR BAYOU PLANT 138 KV line	Rebuild line with higher-capacity conductor	\$1,644,000	3% (\$48,000)
HIGHLAND CITY - CENTRALIA 138 KV	Rebuild line with higher-capacity conductor	\$5,366,000	7% (\$395,000)
CEDAR FALLS WI - OYSTER RIV 138 KV	Rebuild line with higher-capacity conductor	\$2,990,000	10% (\$289,000)
CEDAR BAYOU PLANT - JIM HILL 138 KV L	Rebuild line with higher-capacity conductor	\$3,802,000	3% (\$111,000)
N RANDOLPH - YELLOW CRK 138 KV	Rebuild line with higher-capacity conductor	\$10,715,000	15% (\$1,602,000)
MUKONAGO - PACIFICA 138 KV	Rebuild line with higher-capacity conductor	\$2,820,000	7% (\$207,000)
HIGHLAND CITY - MUKONAGO 138 KV	Rebuild line with higher-capacity conductor	\$20,635,000	7% (\$1,518,000)
N RANDOLPH - BAIROA 3008 138 KV	Rebuild line with higher-capacity conductor	\$8,700,000	15% (\$1,301,000)
WALTHALL - HAGGIN 138 KV	Rebuild line with higher-capacity conductor	\$2,255,000	6% (\$140,000)
SKAGGS ISLAND JUNCTON 2 - MITCHELL DAM 345 KV	Rebuild line with higher-capacity conductor	\$43,351,000	2% (\$1,062,000)
	Total allocated to project		\$11,865,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.